

Machine Learning-Based Polysomnography Data Analysis for ADHD Diagnosis: A Focus on Sleep Stage-Based Biomarkers

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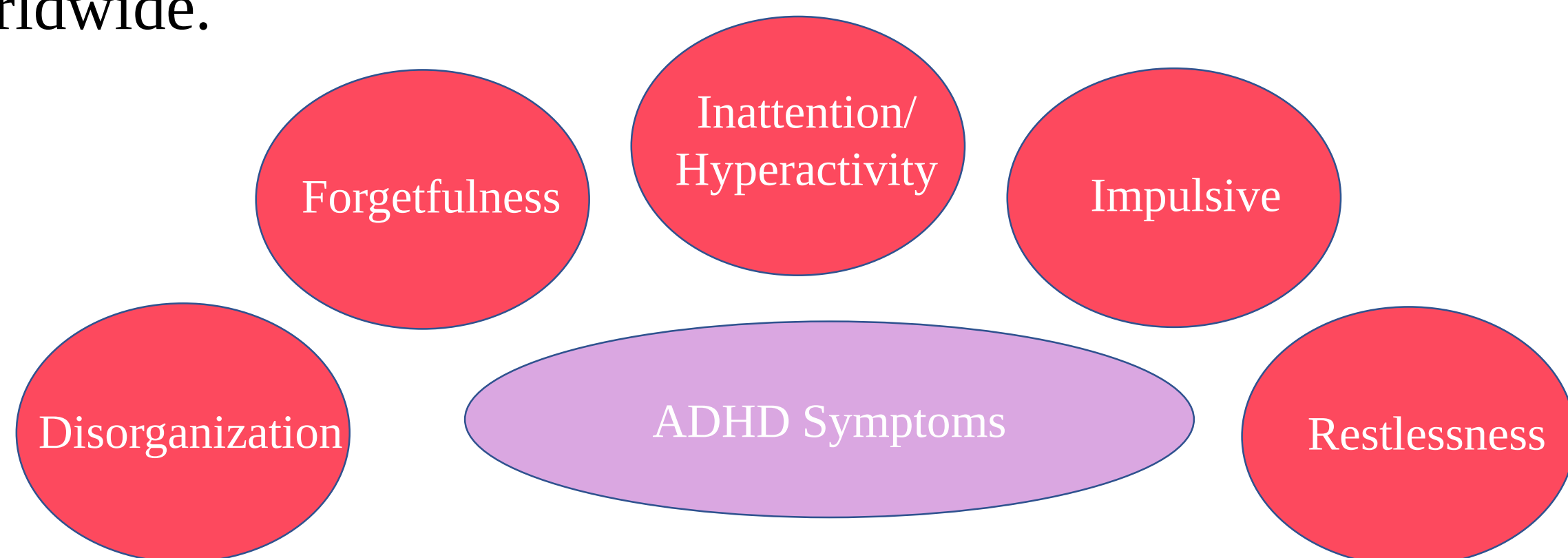
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1. INTRODUCTION

Attention Deficit Hyperactivity Disorder (ADHD) is a **neurodevelopmental disorder** affecting **7.6%** of children, **5.6%** of adolescents, and **6.76%** of adults worldwide.



Current Diagnostic Challenges:

Subjective Assessment	Variability
Misdiagnosis Risk	Delayed Diagnosis

Need for Objective Biomarkers— biological indicators that can reliably diagnose ADHD.

“Can PSG sleep patterns enhance ADHD detection through machine learning and graph theory?”

3. METHODOLOGY

A. Data Acquisition Overview

Aspect	Details
Sample	48 patients (ages 6-18): 25 ADHD, 23 controls
Source	University of Mississippi Medical Center, 2023
PSG Recording	8 hours, 17 channels (2 ECG, 5 EMG, 2 EOG, 8 EEG)
Epochs	30-second epochs, sampled at 512 Hz
Total Epochs	20,294 ADHD, 19,968 non-ADHD
Sleep Stages	Wake, Stage 1, Stage 2, Stage 3-4, REM

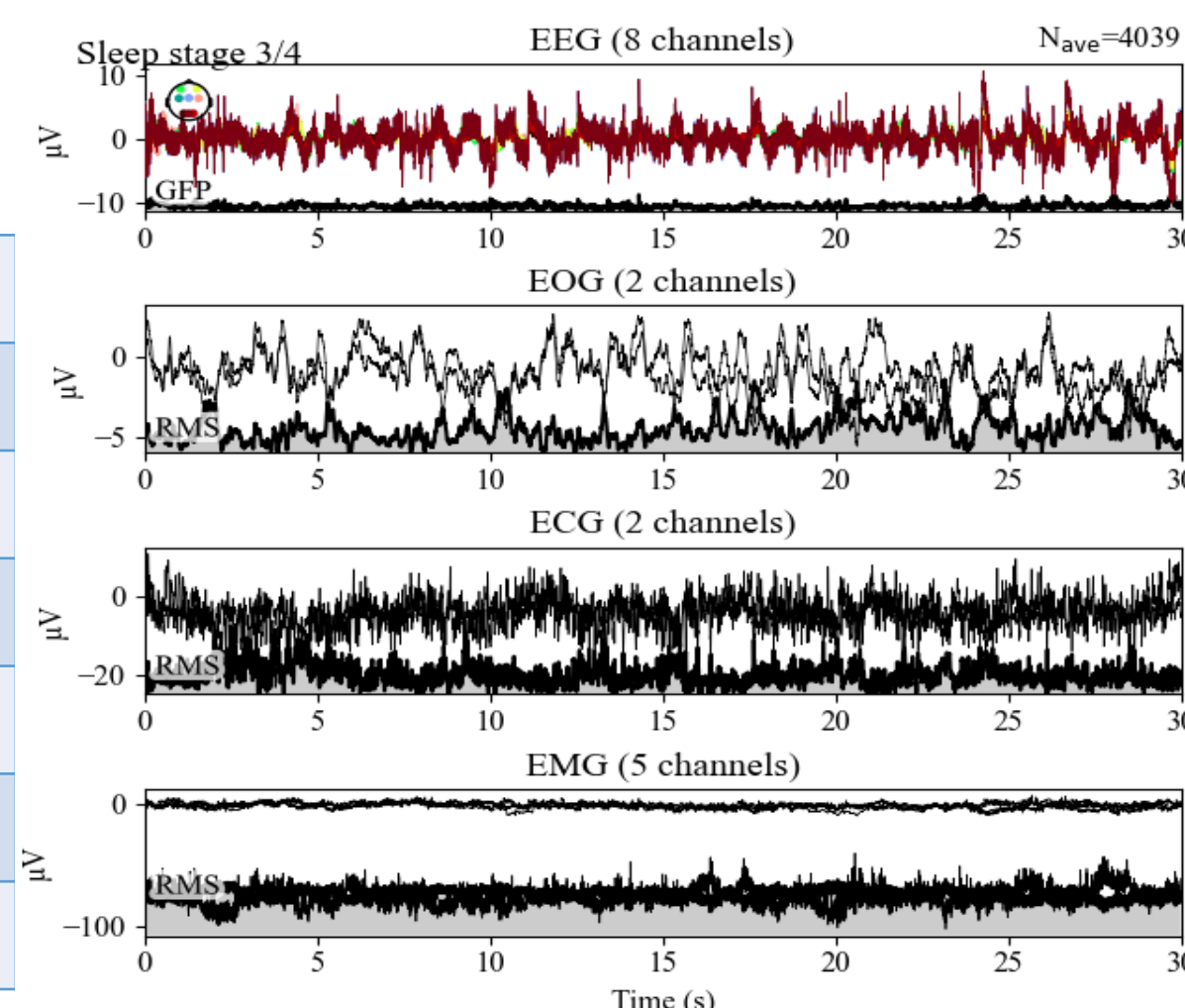


Fig 1: PSG recordings across different channels for stage 3-4 for an ADHD patient

B. Feature Extraction

1. Calculated the **correlation matrices** using **Pearson correlation coefficients**

$$R_{ij}^{(e)} = \frac{\text{cov}(X_i^{(e)}, X_j^{(e)})}{\sigma_{X_i}^{(e)} \cdot \sigma_{X_j}^{(e)}} \rightarrow \begin{array}{l} \text{Covariance between sensors } i \text{ and } j \text{ during epoch } e \\ \text{Standard Deviation} \end{array}$$

3. Extracted the **average shortest path length**

Table 1: A five-dimensional feature vector

Patients	Features (Average shortest path)					Y
	Wake	Stage 1	Stage 2	Stage 3-4	REM	
1	0.101	0.064	0.090	0.086	0.069	1
...	...	...	...	...	...	...
48	0.069	0.063	0.067	0.086	0.055	1

2. Formed the adjacency matrix

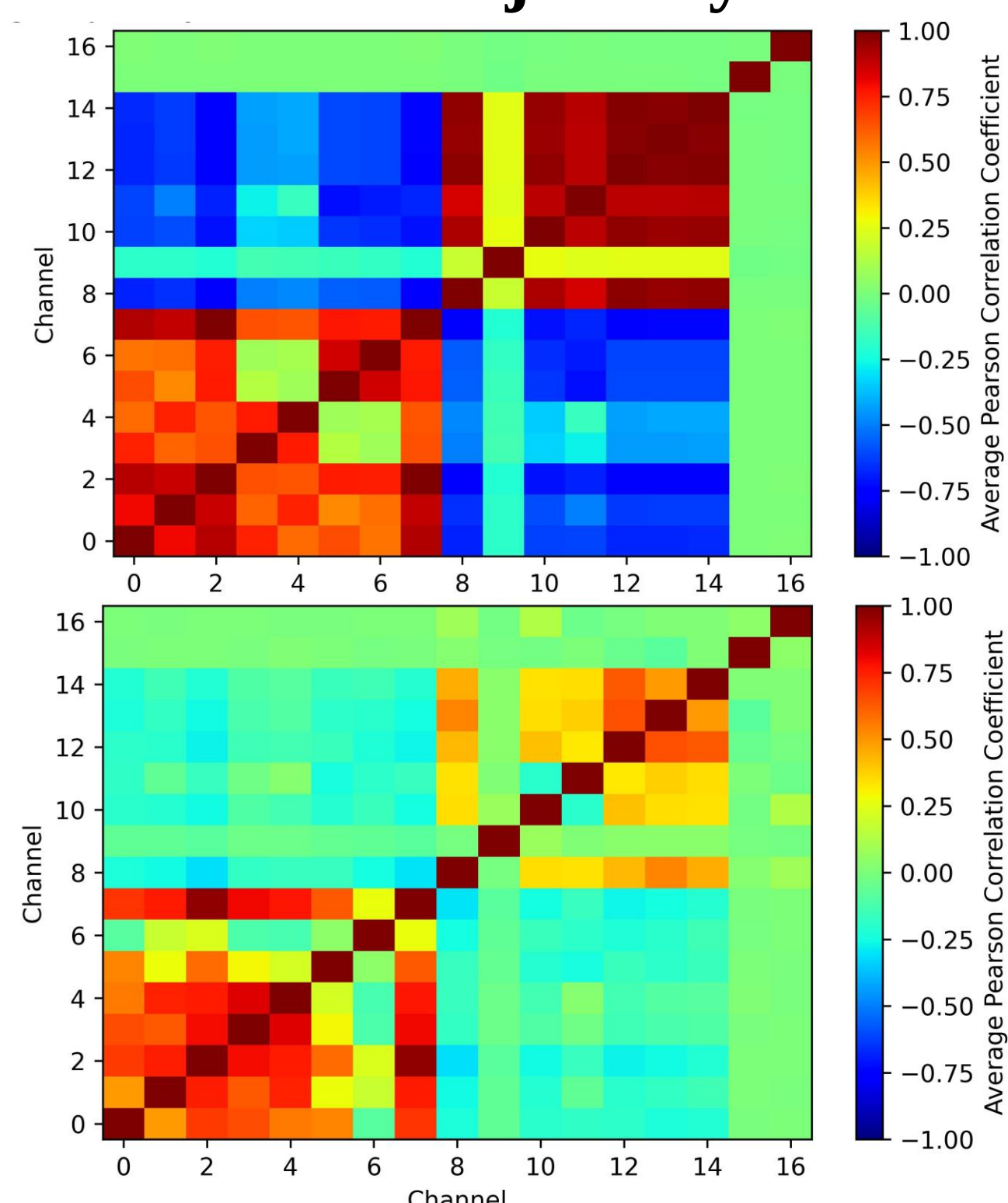


Fig 2: Average adjacency matrix for patient 1 for stage 3-4 (Up) and stage 1 (down)

C. Machine Learning Model

Random Forest classifier to predict ADHD status from 5D feature vectors.

Handel high dimensional data effectively	Robust to overfitting due to its ensemble nature
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5. CONCLUSION & FUTURE WORK

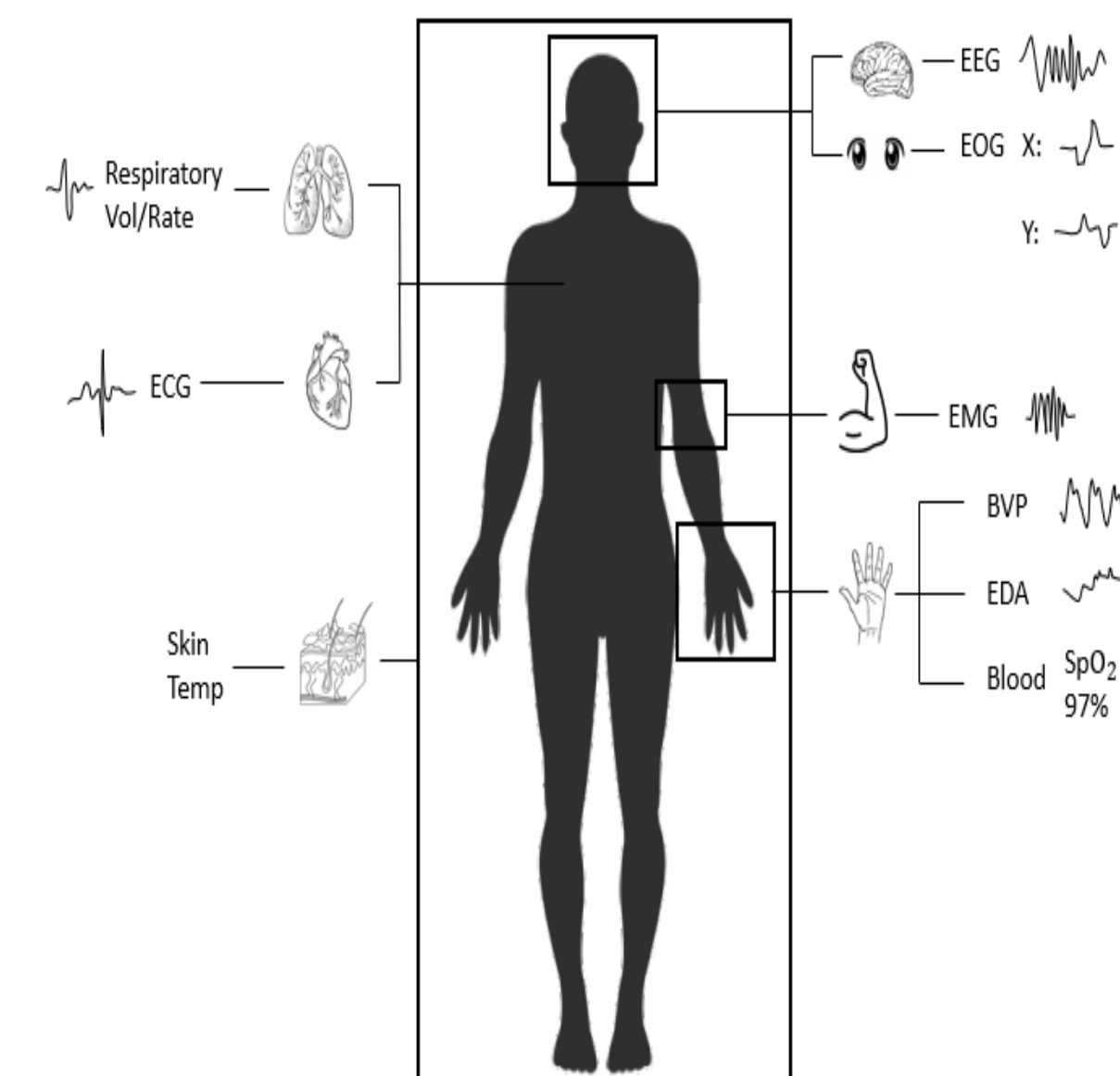
- Enhanced ADHD diagnosis using PSG data across 17 channels, graph metrics (average shortest path length), and Random Forest in 48 patients across 5 sleep stages.
- Achieved 72% accuracy, 71% precision, 85% recall, and 76% F1-score; Sleep Stages 1 and 3-4 most predictive
- ADHD patients showed disrupted transitions (correlation of 0.09, Stage 1 to 3-4) vs. non-ADHD (0.63), indicating impaired light-to-deep sleep progression.
- Suggests sleep fragmentation as an ADHD biomarker, aiding clinical diagnosis.

Future Work:

Compare EEG/EOG/EMG/ECG correlations to identify signal-specific biomarkers for improved accuracy.

2. OBJECTIVES

- Develop a **novel machine learning model to diagnose ADHD** using polysomnography (PSG) data, targeting sleep stage-specific biomarkers.
- Evaluate model performance and robustness for ADHD classification using **comprehensive PSG signals** (EEG, EOG, EMG, ECG).
- Quantify the efficiency of information transfer within brain and other physiological networks **using graph theory metrics**, focusing on the **average shortest path length** across sleep stages.



4. RESULTS

A. Model Performance

Table 2: Evaluation metrics of the classifier for ADHD prediction based on 5-fold cross-validation

Fold	Accuracy (%)	Precision(%)	Recall(%)	F1-Score (%)
1	70	75	75	75
2	68	62	89	73
3	63	67	83	74
4	68	54	100	70
5	89	100	80	89
Average	72	71	85	76

B. Feature Importance Analysis and Correlation Analysis

Figure 3 shows that **Sleep Stages 1 and 3-4** have the **highest importance** scores, highlighting their key role in the model's performance.

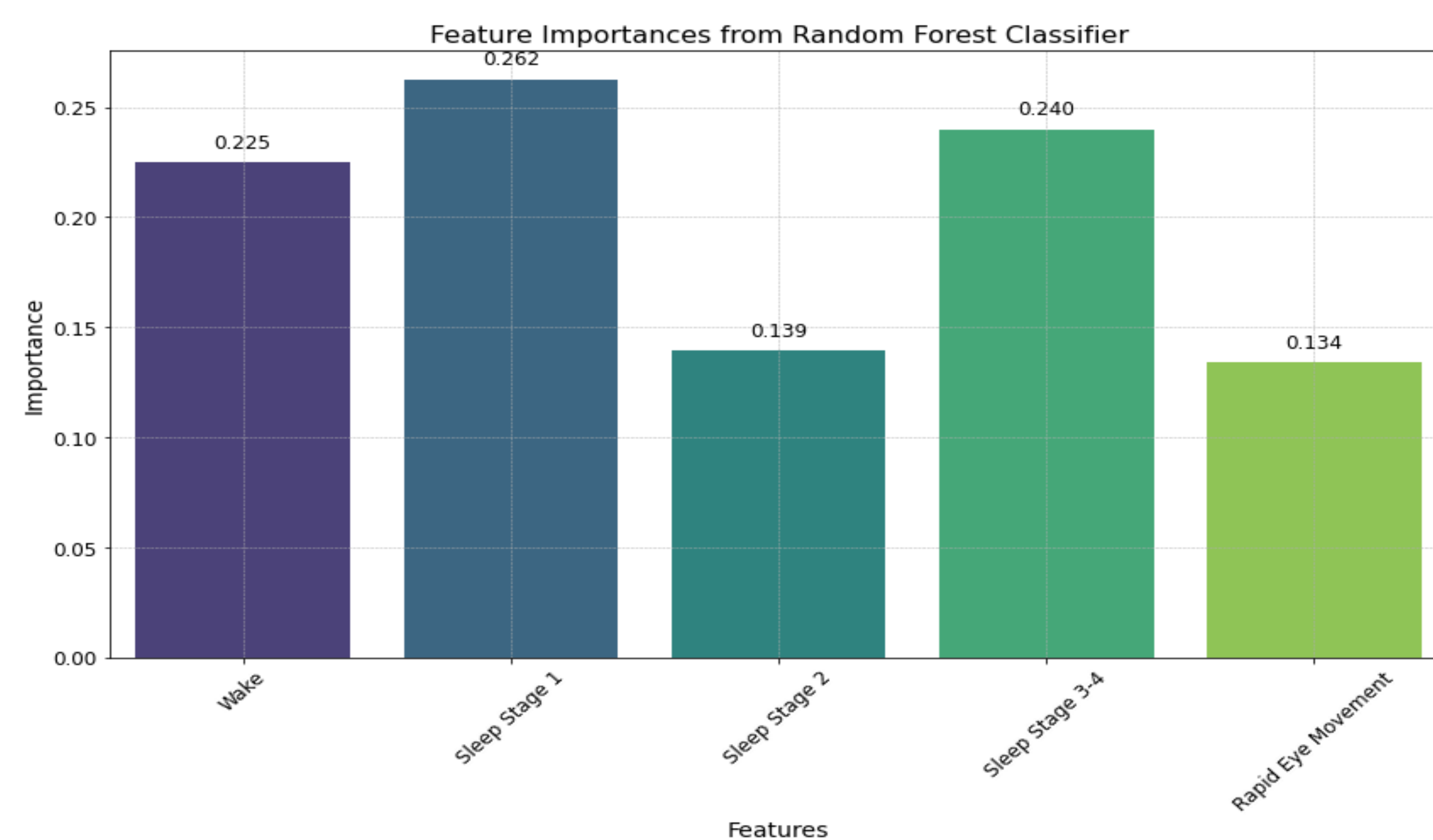


Fig 3: Importance of average shortest path lengths across sleep stages for ADHD prediction.

Correlation analysis (Table 3) compares sleep stage transitions between ADHD and non-ADHD groups. **Higher correlation indicates smoother transitions**, while **lower values suggest fragmented sleep patterns**.

Table 3: Correlation coefficient matrices between different sleep stages for non-ADHD (underlined) and ADHD individuals

Stage	Wake	Stage 1	Stage 2	Stage 3-4	REM
Wake	1.00	0.61	0.63	0.48	0.59
Sleep stage 1	<u>0.59</u>	1.00	0.21	0.09	0.58
Sleep Stage 2	<u>0.71</u>	0.82	1.00	0.75	0.35
Sleep Stage 3-4	<u>0.47</u>	0.63	<u>0.81</u>	1.00	0.10
REM	<u>0.57</u>	<u>0.61</u>	<u>0.73</u>	0.57	1.00

6. SELECTED REFERENCES

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